

AMRUTA NAKATE

San Francisco, CA | (312) 866-4243 | amusnakate97@gmail.com |  amruta-nakate

SUMMARY

Senior Data Engineer with 4+ years of experience building scalable data pipelines, Spark-based ETL workflows, and analytics platforms at Visa. Skilled in distributed data processing, AWS, Snowflake, and developing reliable data solutions that enable business analytics and AI-driven applications. Strong software engineering foundation with expertise in data structures, algorithms, and system design.

EXPERIENCE

Visa Inc: California, USA

Oct 2023 - Oct 2026

Data Scientist (Data Engineering)

- Designed and implemented scalable Spark/Scala ETL pipelines to transform B2B data ingested from 4+ enterprise systems into analytics-ready datasets, applying business logic to create reporting tables that processed millions of records and powered 5+ business-critical dashboards.
- Integrated multiple B2B data sources into an existing segmentation platform by developing robust ETL workflows to enable self-service segment creation and analytics capabilities at scale.
- Designed an AI-driven conversational interface that enabled users to define target segments using natural language, automating segment creation workflows through integration with existing data platform services.

Visa, Inc: California, USA

Sept 2022 - Oct 2023

Senior Data Engineer

- Developed and optimized Spark/Scala ETL pipelines to generate campaign engagement and transaction metrics datasets, processing 1TB of historical transaction data and campaign response events while improving scalability and execution efficiency through Spark performance tuning.
- Designed and implemented scalable data pipelines for an automated scoring framework that combined consumer- and card-level (PAN) insights, generating behavioral scores to enhance segmentation capabilities and support analytics-driven audience creation.
- Implemented and maintained Airflow DAGs for scheduling and orchestrating data pipelines, supporting around 50+ reliable production workflows.

Coupa: Cincinnati, USA

May 2022 - July 2022

Machine Learning Software Engineer Intern

- Optimized K-means clustering models for item similarity analysis through feature engineering, selection, and model redesign, improving cluster quality by 20–30% based on silhouette score and enhancing intra-cluster cohesion.
- Developed a PCA-based dimensionality reduction approach, reducing feature space by 40–60% while retaining 85–95% explained variance, improving clustering stability and interpretability.

Persistent Systems: Pune, India

July 2019 - Apr 2021

Software Engineer

- Developed AWS Glue ETL pipelines to process manufacturing data and built Tableau dashboards for production quality analytics, enabling identification of defect patterns and supporting reduction of machine part defects by 70%.
- Built Python-based data validation frameworks to verify data accuracy and consistency during Snowflake migration, improving reliability of migrated datasets.

TECHNICAL SKILLS

- **Programming:** Python, Scala, SQL, Java
- **Data Engineering:** Apache Spark, PySpark, Hadoop, Kafka, Airflow, AWS Glue
- **Data Science & ML:** Pandas, NumPy, Scikit-learn, TensorFlow, Matplotlib, Seaborn, Generative AI, LLM Applications
- **Databases:** MySQL, MongoDB, Cassandra
- **Data Analysis:** Tableau, MS-Excel, Power BI

EDUCATION

University of Cincinnati: OH, USA

Aug 2021 - Aug 2022

- Master of Engineering in Computer Science - CGPA : 3.8 / 4

University of Pune: India

Jul 2015 - July 2019

- Bachelor of Engineering in Computer Engineering - CGPA : 8.43 / 10